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#include <WiFi.h>
#include <ThingSpeak.h>

const char* ssid = "Wokwi-GUEST";
const char* password = "";

WiFiClient client;

unsigned long channelNumber = 3417056;
const char* writeAPIKey = "P4D3G10ARP7FE69L";

const int sensorPin = 36;    // VP (GPIO36)
const int ledPin = 21;

void setup() {
    Serial.begin(115200);

    pinMode(ledPin, OUTPUT);

    WiFi.begin(ssid, password);

    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }

    Serial.println("\nWiFi Connected!");

    ThingSpeak.begin(client);
}

void loop() {

    int lightValue = analogRead(sensorPin);

    Serial.print("Light Intensity: ");
    Serial.println(lightValue);

    // LED Control
    if (lightValue < 2000) {
        digitalWrite(ledPin, HIGH);
    } else {
        digitalWrite(ledPin, LOW);
    }

    ThingSpeak.setField(1, lightValue);

    int status = ThingSpeak.writeFields(channelNumber, writeAPIKey);

    if (status == 200) {
        Serial.println("Data sent to ThingSpeak.");
    } else {
        Serial.print("Error: ");
        Serial.println(status);
    }

    delay(20000);    // ThingSpeak allows updates every 15 seconds
}

```